

CHUNCHEON, Korea, Republic of

WMO: 471010

Lat: 37.9025N

Lon: 127.7358E

Elev: 258

StdP: 14.56

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.4	9.2	-7.9	3.6	15.6	-4.0	4.5	17.1	14.5	27.6	12.2	26.4	1.7	0	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.1	91.2	76.4	88.7	75.0	86.4	73.5	79.7	86.3	78.3	84.6	77.0	82.8	4.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.2	147.7	82.5	76.6	140.1	81.4	75.3	133.9	80.4	43.6	86.4	42.1	84.9	40.8	82.8	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.8	9.9	8.5	DB	0.1	94.6	5.1	2.1	-3.5	96.1	-6.5	97.4	-9.4	98.6	-13.0	100.1	
			WB	-0.7	82.0	4.9	2.6	-4.2	83.8	-7.1	85.3	-9.9	86.7	-13.4	88.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	25.0	30.8	41.2	53.3	63.9	72.4	77.5	78.0	68.8	56.8	42.5	29.0
	DBStd	19.44	7.36	6.92	6.41	6.69	4.95	4.19	4.12	4.54	4.89	6.32	7.54	7.41
	HDD50	2551	774	539	281	41	0	0	0	0	0	19	245	652
	HDD65	5442	1239	959	736	353	80	4	0	0	19	259	676	1117
	CDD50	3786	0	0	9	140	432	671	854	867	563	230	20	0
	CDD65	1205	0	0	0	3	47	225	389	403	133	5	0	0
	CDH74	9548	0	0	0	77	542	1713	3125	3385	687	19	0	0
	CDH80	2996	0	0	0	8	101	527	1038	1210	112	0	0	0
Wind	WSAvg	2.8	2.5	2.7	3.5	3.6	3.2	2.8	2.6	2.6	2.5	2.2	2.4	2.4
Precipitation	PrecAvg	52.1	0.7	1.0	1.5	3.0	3.8	5.1	15.0	11.7	5.7	1.9	1.8	0.9
	PrecMax	81.5	3.2	3.8	4.3	8.7	11.1	18.1	36.7	30.9	19.4	5.4	5.6	3.1
	PrecMin	26.6	0.0	0.0	0.2	0.1	0.5	0.8	4.6	1.3	0.2	0.1	0.3	0.1
	PrecStd	11.9	0.7	0.9	1.0	2.0	2.2	3.8	8.4	6.8	4.8	1.3	1.3	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	54.9	67.9	80.8	86.5	91.3	94.1	94.4	86.2	76.4	65.4	49.6
		MCWB	38.7	42.6	51.8	59.5	63.7	69.6	78.7	78.3	72.6	62.7	54.3	42.4
	2%	DB	41.9	50.0	62.9	75.9	82.7	88.1	91.0	91.6	83.2	73.0	61.1	45.3
		MCWB	35.2	40.2	48.6	57.3	62.3	69.0	77.0	77.8	70.1	61.6	52.3	39.4
	5%	DB	38.8	46.0	58.4	71.8	80.0	85.6	88.3	89.3	80.7	69.9	57.7	42.6
		MCWB	33.0	37.6	45.8	55.0	61.7	68.4	75.7	77.1	68.6	59.7	50.5	37.6
	10%	DB	36.0	42.2	54.5	68.1	76.7	83.0	85.7	86.7	78.1	67.1	54.3	39.7
		MCWB	31.3	35.1	43.4	52.9	60.3	67.6	74.9	76.0	67.4	58.0	48.2	35.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.0	44.9	55.4	64.7	68.8	75.3	81.8	82.2	76.1	67.4	57.7	44.0
		MCDB	43.9	50.8	63.5	75.0	77.4	83.5	88.1	87.8	82.5	72.3	61.9	47.2
	2%	WB	36.4	41.4	50.5	60.4	66.3	73.1	79.6	80.2	73.6	64.1	54.4	40.5
		MCDB	40.2	48.7	59.2	70.3	75.2	79.8	86.0	87.0	78.9	69.3	59.1	43.9
	5%	WB	34.1	38.3	47.5	57.5	64.5	71.7	78.2	78.9	71.6	61.8	51.7	38.3
		MCDB	37.1	44.3	56.3	67.1	74.1	78.9	84.3	85.4	76.4	67.1	55.7	41.8
	10%	WB	32.2	35.8	44.5	54.8	62.9	70.3	76.9	77.6	69.8	59.8	49.3	35.9
		MCDB	35.4	41.2	53.1	64.7	72.4	78.3	82.4	83.8	75.4	65.2	53.4	39.1
Mean Daily Temperature Range	5% DB	MDBR	18.3	20.0	21.1	23.7	22.0	18.5	13.0	14.1	17.0	19.7	17.5	16.7
		MCDBR	18.5	24.1	27.8	30.5	28.2	23.9	17.9	17.9	19.4	21.1	21.4	17.7
		MCWBR	12.7	14.9	15.2	13.9	10.4	7.9	6.3	6.3	7.7	10.2	12.6	12.2
	5% WB	MCDBR	15.6	21.4	24.9	24.8	22.0	17.1	14.4	15.2	14.8	17.3	17.3	16.3
		MCWBR	11.6	14.3	15.1	13.4	10.2	7.4	6.3	6.4	7.1	9.8	11.4	12.1
Clear-Sky Solar Irradiance	taub	0.419	0.481	0.573	0.623	0.629	0.693	0.662	0.574	0.497	0.491	0.466	0.416	
	taud	1.925	1.820	1.691	1.641	1.668	1.603	1.721	1.930	2.063	1.954	1.951	1.975	
	Ebn at Noon	225	226	219	218	220	207	212	228	239	222	208	215	
	Edh at Noon	44	56	71	79	78	83	73	58	48	49	43	40	
All-Sky Solar Radiation	RadAvg	779	1033	1313	1547	1737	1623	1263	1316	1262	1106	765	695	
	RadStd	58	82	136	125	161	143	185	180	149	83	86	58	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	+108
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+178	+95

Nomenclature: See separate page